PAM-3 decoding
Patent reconstruction and simulation

US 11,405,248 B2 - Prabhakar, Mulki and Mahammad (2022)

A two-channel functional model, a numerical stress suite and independent SystemVerilog verification. Prepared for Alborz Seifaei.

The supplied patent was read in full: 19 PDF pages, 12 drawing sheets and 19 claims. Published rules are retained; missing implementation details are explicitly reconstructed.

What the executed tests establish

Executed condition	Failed intervals / evaluated	Interval SER
SNR 30 dB	0 / 40,560	0.0000%
SNR 18 dB	224 / 40,560	0.5523%
SNR 10 dB	8,775 / 40,560	21.6346%
Clock +1000 ppm	0 / 8,112	0.0000%
Literal DC at 30 dB	26,487 / 40,560	65.3033%

The fixed reconstruction works well at high SNR and with clock mismatch in the tested synthetic channels. It is not uniformly best: the simple fixed-clock baseline wins several noise-only tests. The inferred adaptive law helps some offset/gain cases and hurts others.

The literal DC equation produces a conditional conflict with the negative-level threshold under the stated interpretation. This is a disclosure/reconstruction issue, not a verified defect in the vendor's physical product.

Scope: no FPGA-board experiment, no Vivado execution, no synthesis/timing closure at 800 MS/s, no measured coupler model and no complete Ethernet frame decoder.

1. Architecture and published parameters

The patent probes a full-duplex automotive Ethernet line. A directional coupler separates both directions; two DVGAs set amplitude, two ADCs acquire samples and two FPGA paths decode PAM-3 symbols. USB passes symbols to host software for higher-layer decoding.

Processing stage	Implemented rule
ADC interface	Behavioural already-deserialized 8-bit ADC codes; no ISERDES electrical model
DC correction	Causal 128-sample MAX/MIN window, optional literal equation or disclosed bypass
Slope / level decoder	Four thresholds: Plus 140, Minus 95, ZeroPlus 160, ZeroMinus 80
Spike filter	Reconstructed removal of isolated <=2-sample runs with equal neighbours
Must transition / midpoint	Opposite edge directions within <=12 samples; midpoint inside bounded plateau
Edge filter 1 / 2	Keep must or >=9-sample edge distance; reject edge before a must edge within 4
Sampling / output	12-sample free-run corrected by must midpoints; retain must points, remove near future duplicates

Nominal sampling is 800 MS/s, or 1.25 ns/sample; 12 samples/symbol imply approximately 66.667 MBd. The patent's '66.66 Mbps' wording is not used as the symbol-rate unit here. Extension to 1000BASE-T1 is claimed in the patent but not fully specified or implemented in this package.

Nominal ADC levels 60/127/194, initial receiver phase 6, the DC guard value 80, and integer midpoint rounding are assumptions. The original document gives threshold codes but not a complete analog calibration or bill of materials.

2. Ambiguities that affect reproduction

DC correction

Column 6 prints y = x - MIN + Minus_Threshold, conditioned on MAX, MIN and y <255. With the same Minus_Threshold=95 used for slicing, every corrected value satisfies y >=95, because the causal window includes x. The mirrored strict falling rule requires y <95 to decode -1. Active correction and that rule cannot coincide.

The reconstruction tests the equation literally and also tests the patent's allowed DC bypass. It does not silently repair the equation. The unknown guard is varied separately; changed anchor values are labelled engineering variants. Separate register meanings or unreported equality handling could resolve the issue in real hardware.

Must transition

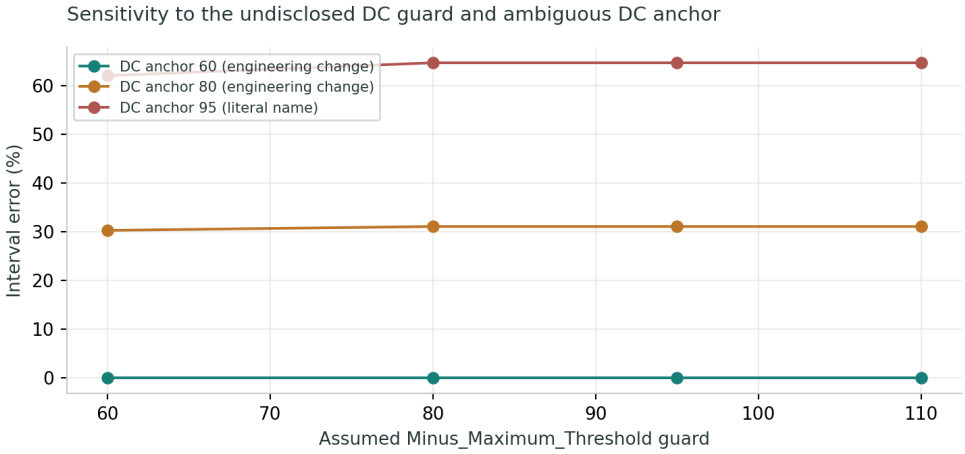
Figure 2 refers to opposite slopes, and Figure 4G shows a three-sample zero valley between +1 levels. The implemented interpretation pairs opposite edge directions, rather than requiring a +1-to--1 level change. The text is ambiguous; this is a documented design decision.

Spike filtering and time recovery

Figure 4F illustrates spike removal but gives no complete state machine. The chosen <=2-sample isolated-run rule preserves Figure 4G's three-sample pulse. The patent does not specify startup phase, midpoint rounding, or all edge-distance bookkeeping details. Frame-based lookahead is used in the complete HDL reference; bounded delays and buffered histories would be required for a streaming realization.

Adaptive thresholds

The patent gives a 12-sample default update block but no exact update equation. The optional causal decision-directed EWMA here uses previous decoded classes, rate 1/8 and >=3 samples per class. It receives no TX labels, SNR, gain or DC truth. This law is an engineering hypothesis, not the vendor's algorithm. Incorrect class assignments can reinforce bias.



3. Channel, evaluation and reproducibility

The main suite generated 528,384 channel-symbol exposures and executed 516 method/channel trials. There are seven SNR points, five independent seeds and two channels per SNR, plus 17 stress cases and a 5 x 5 gain/offset grid. Phase and DC sensitivities are additional runs.

Symbols are synthetic random or structured PAM-3, not an IEEE-compliant PCS-encoded Ethernet stream. A first-order low-pass models finite edges; echo is one delayed tap; reverse-channel leakage is an independent PAM-3 contributor. A DVGA gain/offset and a saturating rounded 8-bit ADC complete the behavioural path.

Noise variance uses nominal transmitted zero-mean symbol power: 67^2 E[s^2] / 10^(SNR/10). Gain, echo and clipping are applied separately; a changed gain changes effective received SNR. Clock offset changes TX symbol duration relative to the ADC. Independent boundary jitter is clipped to prevent invalid durations.

Evaluation without oracle-assisted decoding

TX timing and labels are used only after decoding. Each original TX interval is correct only with exactly one receiver output inside it and the right value. Missing and duplicate intervals are failures. No edit-distance rematching or best-phase selection hides synchronization loss. The first 32 and final 8 symbols are excluded by a fixed declared policy.

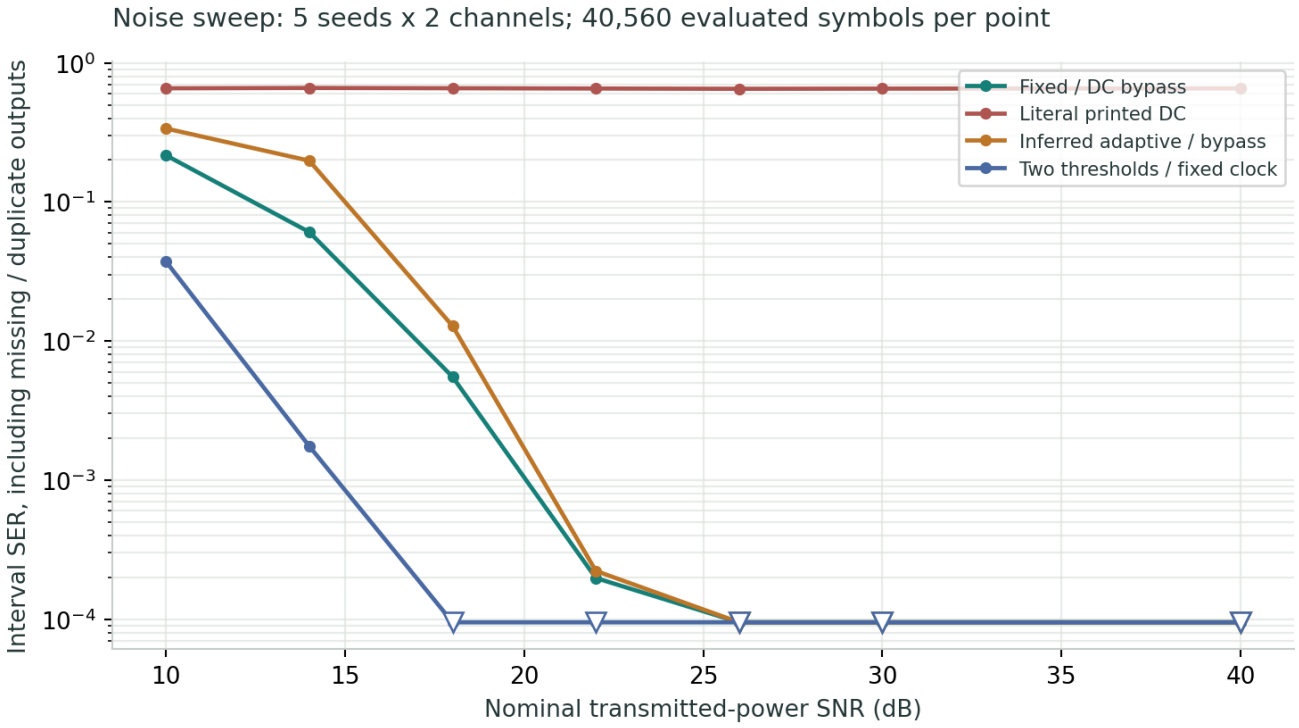
Interval SER = failed TX intervals / evaluated TX intervals. CSV files also retain missing intervals, duplicate intervals, extra outputs, amplitude errors among singly represented intervals, clipping and RMS/P95 timing position relative to the true symbol center.

Statistics

Every SNR/method point pools 40,560 evaluated symbols. Wilson 95% bounds are descriptive; time-correlated burst errors violate independent-binomial assumptions. Independent-run min/max and SD are retained. Zero observed errors gives a finite upper bound and does not prove zero error probability. Stress/grid trials have fewer independent seeds and are exploratory, not field reliability estimates.

Seeds, configuration, Python/NumPy versions and run duration are in results/metadata.json. All outputs are regenerated from executable simulation code. Reproduction also requires SciPy; current library versions are recorded in the environment snapshot included with the package.

4. Noise sweep

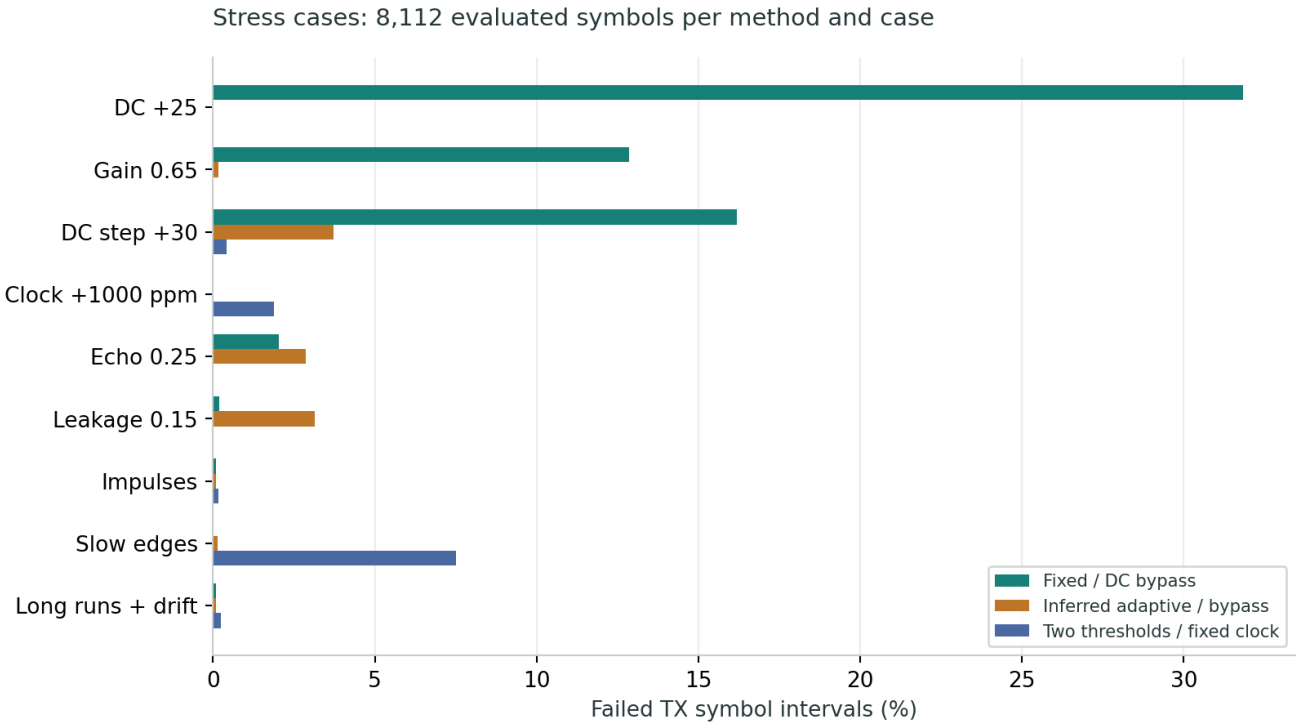


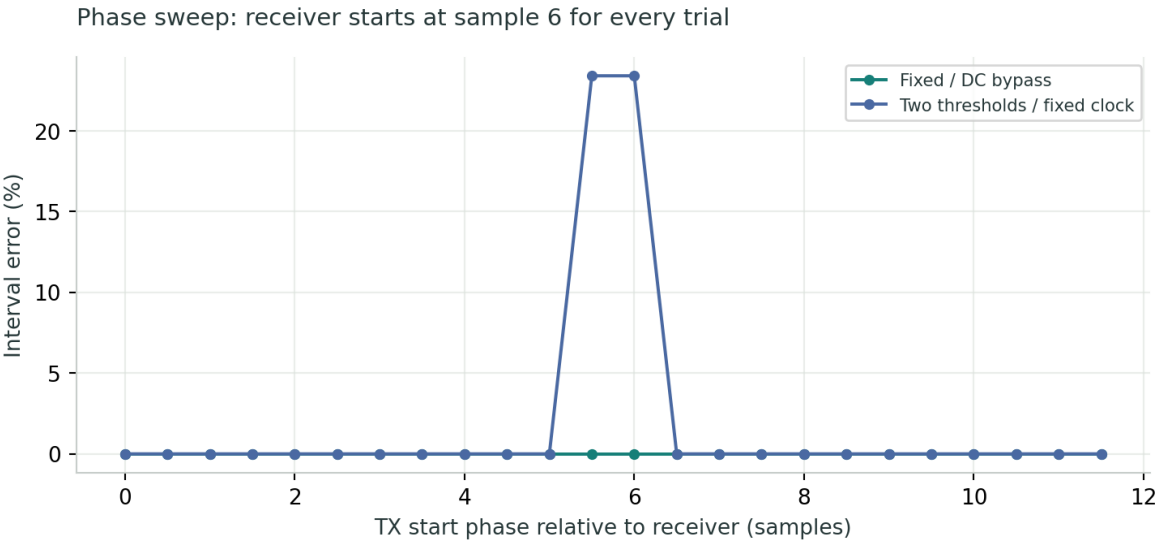
Downward hollow markers: zero observed errors plotted at the descriptive 95% Wilson upper bound.

SNR (dB)	Fixed bypass (%)	Adaptive (%)	Simple baseline (%)
10	21.6346	33.7451	3.7229
14	6.0700	19.7880	0.1750
18	0.5523	1.2821	0.0000
22	0.0197	0.0222	0.0000
26	0.0000	0.0000	0.0000
30	0.0000	0.0000	0.0000
40	0.0000	0.0000	0.0000

In this noise-only sweep the fixed-clock baseline retains a favourable sampling phase and has no clock drift. It avoids several false timing anchors produced by the reconstructed slope logic. This comparison does not establish that a fixed-clock comparator is generally superior; the independent phase/drift cases test the capability that it lacks.

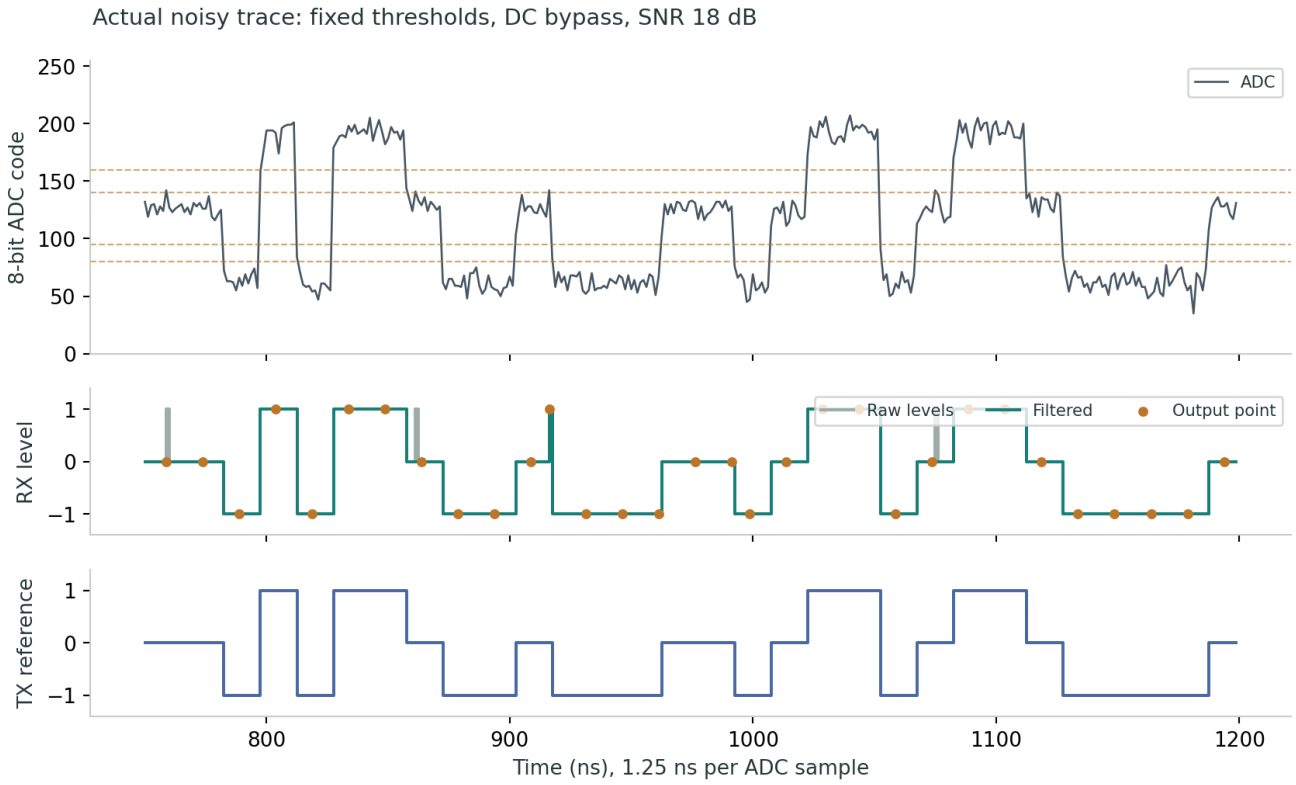
5. Stress, timing and trace results

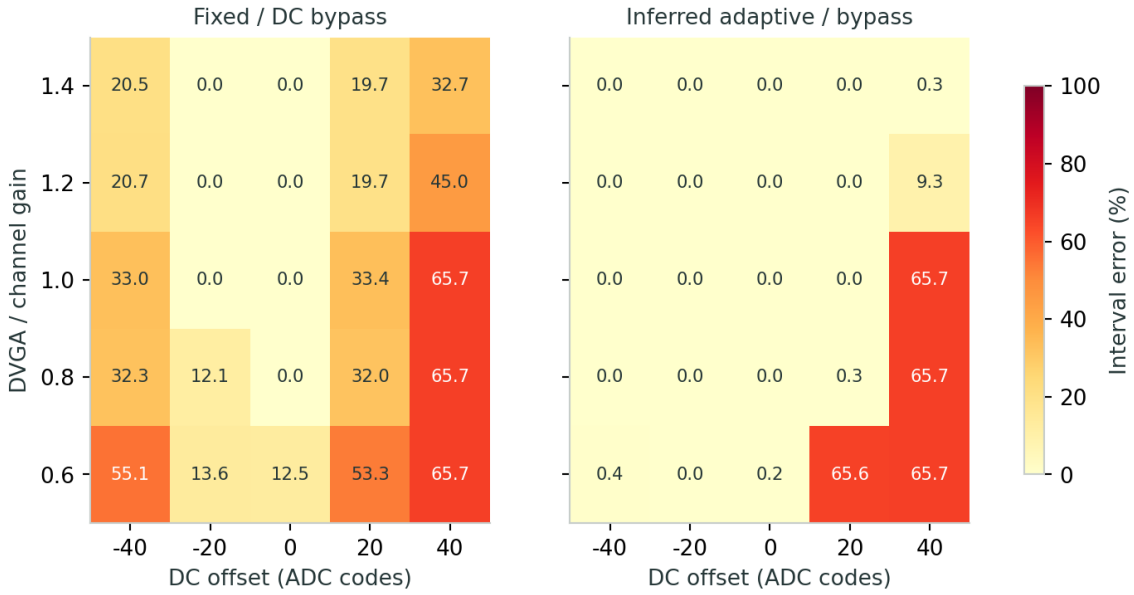




Clock drift is recovered well when useful opposite-edge pairs occur. Long identical runs can accumulate drift until transitions return. Slow-edge and impulse scenarios also illustrate why a timing/filter chain can help. Gain/DC asymmetry and false timing anchors remain important limitations.

6. Actual waveform and calibration sensitivity





The waveform is a real stored trace from the executed model. TX reference is displayed for analysis only and is not passed to the decoder. The offset/gain map is exploratory (2,008 evaluated symbols/channel per cell); it does not demonstrate universally robust calibration.

7. Independent HDL verification

Icarus Verilog 12.0 executed 12 dual-channel cases and compared 221,880 channel samples. Every case passed. Scenarios: clean, SNR 10 dB, clock drift, impulses, DC step and long runs, each with DC bypass and literal DC enabled.

Case	DC bypass	Literal DC
clean	PASS	PASS
noise10	PASS	PASS
drift	PASS	PASS
impulses	PASS	PASS
step	PASS	PASS
long_runs	PASS	PASS

Checked quantities

Every ADC sample's corrected value and raw level; complete symbol output counts; every output sample index and symbol value for both channels. Front-end input-valid stalls were deliberately inserted to check state retention. Comparison vectors and logs are included.

Additional validation: 17 Python/model/API tests passed. A DOM integration test exercised saved-waveform controls, table filters and live-result application without script errors. Canvas calls were mocked in that test; native browser rendering was blocked by this execution environment and is not claimed. The PDF itself was rendered and inspected.

What the files represent

pam3_reference.sv is a complete frame-based functional HDL reference; it is simulation-only. pam3_frontend.sv is a synthesizable DC/slope implementation, but its large unpipelined extrema reduction is not a production 800 MS/s architecture. The optional inferred adaptive law is Python-only. tb_dual.sv generates a 100 MHz testbench clock; its VCD is not a throughput measurement.

Vivado script

A Tcl script is supplied to run a saved comparison case in Vivado/XSim. It was not executed here. No post-synthesis or post-route check, target FPGA selection, resource/power report, ISERDES capture, USB transfer or hardware experiment was performed.

Next hardware milestones

Select compatible ADC/FPGA interfaces; define sample-parallel processing and delay budgets; design and pipeline the full timing chain; synthesize and close timing; validate real coupler/DVGA/ADC models; acquire real bidirectional traffic; implement USB/host PCS decoding; compare against known traffic and measure losses.

8. Full stress summary

Scenario	Fixed bypass %	Inferred adaptive %	Simple baseline %
ideal	0.0000	0.0000	0.0000
offset_plus25	31.8417	0.0123	0.0000
offset_minus25	0.0000	0.0000	0.0000
gain_065	12.8452	0.1603	0.0000
gain_135	0.0000	0.0000	0.0000
dc_step_plus30	16.1859	3.7352	0.4191
gain_step_minus035	6.8047	0.1356	0.0000
clock_plus1000ppm	0.0000	0.0000	1.8861
clock_minus1000ppm	0.0000	0.0000	1.6765
jitter_08samples	0.0000	0.0000	0.0370
echo_025	2.0340	2.8723	0.0000
coupler_leakage_015	0.1972	3.1558	0.0000
impulses	0.0986	0.0986	0.1603
slow_edges	0.0000	0.1479	7.5074
long_runs_clock_drift	0.0986	0.0986	0.2465
only_minus_zero	0.0000	0.0000	0.0000
alternating	0.0000	0.0000	0.0000

All literal-DC results, per-run variation, missing/extra counts and timing metrics are in summary.csv and all_runs.csv. No failed condition has been removed from the delivered suite.

9. Open, reproduce and extend

Open PAM3_Results.html directly to browse saved plots, method comparisons and full-resolution trace windows offline. Use app.py for an actual new calculation with changed channel settings; viewing saved trace controls alone does not rerun the model.

Task	Command from extracted package directory
Install Python packages	python -m pip install -r requirements.txt
Run meaningful tests	python -m unittest discover -s tests -v
Repeat numerical suite	python run_simulation.py --symbols 4096 --repetitions 5
Repeat HDL check	python verify_hdl.py (Icarus Verilog required on PATH)
Rebuild plots and report	python make_report.py
Interactive local experiments	python app.py; open http://127.0.0.1:8765
Decode measured ADC CSV	python decode_csv.py measured.csv --out decoded

Windows launchers install dependencies in a local .venv. No API key or cloud service is used. The generated results are already supplied, so opening them needs neither Python nor installation.

Source and implementation references

Primary method source: supplied US11405248B2 PDF, especially Figures 1-5, columns 5-8 and claims 1-19. Exact assumptions and optional update equations: docs/ASSUMPTIONS.md. Model implementation: pam3/model.py; channel/evaluator: pam3/channel.py.

Tool documentation: steveicarus.github.io/iverilog/usage/ (Icarus Verilog); amd.com/en/products/software/adaptive-socs-and-fpgas/vivado/verification.html (Vivado simulation). No external experimental paper results were used in the numerical report.

Use in research: call these results an assumption-labelled functional reconstruction. Do not attribute the inferred adaptive law, synthetic channel parameters, or measured error rates to the patented commercial product.